The Fractal Predictive Cortex: a training-free, alignment-free, self-organising model that unifies prediction, anomaly detection, classification, and phylogenetics for biological sequences

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Working draft — a reproducible methods paper. Every number and figure is produced by the accompanying open-source code (`benchmarks/` and `examples/`; see §6) and is reported honestly, including the several cases where the method only matches, or is beaten by, its baselines.

Abstract

Alignment-free sequence analysis usually means picking a task (compression, classification, distance estimation, anomaly detection) and then picking a tool built specifically for it, each with its own hyperparameters — most importantly the k-mer length *k*. We present the **Fractal Predictive Cortex (FPC)**, a single self-organising model that performs all four tasks at once, with **no training labels, no sequence alignment, no pretrained weights, and no fixed k-mer order**. FPC streams a sequence through a Chaos Game Representation, which turns "the recent context" into "a point in the plane", and grows a cloud of predictive **prototypes** wherever it is *surprised* — where its own next-symbol prediction fails. Its single objective, cumulative surprise, is exactly the sequence's description length (the Minimum Description Length principle), and the four capabilities are four read-outs of that one quantity. On controlled data with known ground truth, FPC (i) beats a general-purpose compressor (gzip) and low-order Markov models and remains competitive with the *best hand-tuned* fixed-order Markov model **without having to choose an order**; (ii) classifies short reads (down to 150 bp) at 100% where k-mer + kNN/logistic baselines fall to 94–97%; (iii) recovers evolutionary distances **statistically on par with the standard tools FFP and Mash and the prior-art cross-entropy distances NRC and NCCD** across 25 independent controlled phylogenies (mean Pearson r = 0.93; paired Wilcoxon $p \geq 0.24$ against all four) and **significantly better than a naive k-mer-cosine baseline** ($r = 0.78$, $p < 10^{-6}$); and (iv) detects foreign inserts on par with a strong k-mer-novelty detector while adding calibrated, per-position surprise. On real genomes — seven coronaviruses, a 28-genome five-family study, and 40 real proteins — it recovers the correct closely-related groups unsupervised (SARS pair, Dengue serotypes, Ebola species, HIV/SIV; four of five families as clean clades), works unchanged on the 20-letter protein alphabet, and cleanly flags spliced phiX174 contamination (per-window ROC-AUC = 1.0). On the community-standard test — Robinson–Foulds tree distance on 17 real mammalian mitogenomes — FPC **ties Mash, FFP, and the prior-art NCCD** (normalized RF 0.43), sits just behind NRC (0.36), and beats k-mer cosine (0.71). We report the losses just as plainly: on the hardest *deep*- divergence real families the composition baseline **FFP is clearly better** (family purity 0.82 vs 0.93), and near-incompressible viral DNA limits

compression for every method. Because the model *is* a geometry, it renders directly as an interactive 3-D **cognitive map** of its own reasoning. A systematic literature review confirms that FPC's individual ingredients — CGR, self-organising maps, prediction/compression-based comparison, the least-surprised-model classifier, and the surprise = description-length identity — are all established prior art; the contribution is their **unification into a single surprise-driven *growing* memory in CGR space**, for which we find no near-identical precedent. The core engine is ~1300 lines of NumPy/SciPy and runs fully offline.

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1. Introduction

A great deal of modern bioinformatics is *alignment-free*: rather than aligning sequences base-by-base (which is expensive and assumes homology), we summarise a sequence by the statistics of its short subwords (k-mers) and compute on those [Vinga & Almeida 2003; Zielezinski et al. 2017]. This is fast and general, but it fragments into task-specific pipelines: one tool compresses, another classifies, another builds trees, another scans for anomalies — and almost all of them hinge on a hand-chosen word length k whose best value is unknown in advance and differs from dataset to dataset.

We ask a different question: *is there a single model whose one internal signal naturally answers all of these questions?* Our answer draws on two old, elegant, and (individually) well-understood ideas — coupled, as we argue in §1.1, in a way that has no near-identical precedent:

1. **Chaos Game Representation (CGR)** [Jeffrey 1990] maps a sequence to a trajectory of points in the unit square such that *positions sharing a long common suffix land at nearly the same point*. Geometry becomes context.
2. **Predictive coding / surprise minimisation** — the idea, from both data compression (MDL, PPM, context-tree weighting [Rissanen 1978; Cleary & Witten 1984; Willems et al. 1995]) and theoretical neuroscience [Rao & Ballard 1999; Friston 2010], that a good model of a stream is one that predicts the next token well, and that *prediction error (surprise)* is the signal worth acting on.

FPC places a growing, self-organising set of predictive units **in CGR space** and trains them online by surprise: it spends memory (grows a new unit) exactly where its predictions fail. The result is a model that is simultaneously a predictor, a compressor, a novelty detector, a generative classifier, and a distance function — and, because it lives in a 2-D geometry, a picture.

What this paper claims, and what it does not. We do *not* claim state-of-the-art on any single task: a matched fixed-order Markov model beats FPC on Markov-generated compression, and the composition method FFP beats it on deep-divergence real phylogeny — we report those losses plainly. We *do* claim (a) a genuinely unified framework in which four capabilities fall out of one mechanism and one objective; (b) *competitiveness* with standard alignment-free tools — statistically indistinguishable from FFP and Mash on controlled phylogenies (§3.4) and robust on short reads (§3.3); (c) freedom from k-mer-order selection; and (d) a novel, interpretable 3-D visualisation of the model's internal state and a calibrated per-position surprise signal that single-purpose distance tools do not provide.

1.1 Related work and the precise locus of novelty

Two systematic, adversarially-verified literature reviews (one broad, one focused on the hardest residual claims) establish that **every individual ingredient of FPC is prior art**, and pin down exactly what is not:

- **CGR + neural / self-organising models.** CGR/FCGR has been fed to CNNs (*CGRclust*; Alipour, Hill & Kari, 2024) and combined with a static k-mer "dynamical language" weighting for phylogeny (*CGRWDL*; Wang, Yu & Li, 2024). The closest CGR + self-organising-map work is *SOCGR* (Neme et al., 2008) — but it reorders a **fixed** $2^k \times 2^k$ lattice for topological continuity; it does not grow, predict, or use surprise. FPC shares the CGR substrate with these, not the mechanism.
- **The "least-surprised model" classifier is textbook.** Training one predictor per class and labelling a query by `argmax_i P^i(x)` (equivalently, minimum −log P = least surprise) is the winner-take-all variable-order-Markov classifier of Begleiter, El-Yaniv & Yona (2004), demonstrated on SCOP protein folds. We claim no novelty for this rule.
- **Prediction/compression comparison is mature.** Mixture-of-predictor DNA compressors (*GeCo3*, Silva et al. 2020; *Jarvis*, Pratas et al. 2019) and compression distances (NCD/IBD; La Rosa et al. 2013; Ferragina et al. 2007) embody the same prediction-as-modelling philosophy with fixed-topology mixers, not CGR or growing memory.
- **surprise = code length = MDL** is standard information theory (Rissanen; Cover & Thomas), and the FCGR of order k is provably equivalent to the k-mer frequency vector at that order (He, Kari & Millán-Arias, 2025) — so neither the objective nor the representation is itself novel.
- **Growing neural models on sequences already exist** — for clustering, not prediction. A growing self-organising map has been applied to protein-sequence classification (Ahmad, Alahakoon & Chau, 2010) and a neural gas to DNA k-mer barcodes (Fiannaca et al., 2015). Crucially, these grow units on *vector- quantisation* error (or ART vigilance) to cluster fixed feature vectors — **not** on *next-symbol prediction* error, and not for autoregressive prediction.
- **The cross-entropy distance itself is NOT novel.** Defining a genomic distance as the bits to describe x under a predictor trained on y — i.e. the cross-entropy / relative compression of one sequence's learned model on another, used to build phylogenetic trees — is exactly Pratas, Silva & Pinho's **Normalized Relative Compression** (2018) and **NCCD** (Pratas & Pinho, 2014), and the KL-between-per-sequence-Markov-models trees of Bai (2013) and Chang & Wang (2011). We claim **no novelty** for the distance definition, and we benchmark directly against NRC and NCCD (§3.4, §3.9); our only difference is the *substrate* (a growing predictor rather than a finite-context Markov model).

What is left, and is our contribution. After the corrections above, the defensible novelty is narrow and specific: (a) growth driven by **next-symbol prediction error** ("grow-where-surprised") in an *online, autoregressive* predictor of biological sequences — every prior growing model on sequences grows on quantisation error for clustering, not on predictive surprise; (b) an explicit **predictive-coding framing** of sequence modelling, for which the review found no prior genomic/proteomic application; and (c) the **unification** of these with the CGR substrate so that four capabilities and a 3-D interpretable map arise from one surprise signal — no surveyed method combines two or more of these elements. We therefore frame FPC honestly as a novel **unification/architecture with a prediction-error-driven growth rule**, *not* as a new distance, objective, representation, or classifier, and (per the AFproject no-free-lunch finding) make no claim of universal superiority.

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2. Method

2.1 Notation and the alphabet

A sequence is a string over an alphabet of a canonical symbols (DNA: $a=4$; protein: $a=20$), plus a wildcard for ambiguous characters. Everything below is alphabet-agnostic; only the symbol-to-integer map knows about letters.

2.2 The fractal mind-space (Chaos Game Representation)

Fix a vertex $v(s) \in [0,1]^2$ for each canonical symbol (the four corners of the unit square for DNA; a regular polygon on a circle otherwise). Starting from the centre $p_0 = (½, ½)$, read the sequence and iterate the contraction map

$$p_t = (1 - r) \cdot p_{t-1} + r \cdot v(s_t), \qquad 0 < r < 1.$$

For DNA we use the classical $r = ½$; in general $r = \min(½, \sin(\pi/a))$ keeps the per-symbol sub-copies from overlapping. Because the map is a contraction, the influence of a symbol decays as $(1-r)$ per step, so **p_t is an exponentially-weighted memory of the recent suffix**. Concretely, two positions whose last k symbols agree differ by at most $\sim(1-r)^k$ in position (empirically $0.5^8 \cdot \sqrt{2} \approx 0.0055$ for an 8-symbol shared suffix, versus ~ 0.1 for a different suffix — a >10× separation; see `tests/test_core.py::test_cgr_shared_suffix_are_close`). We call $C_t = p_{t-1}$ the **context point** for position t: the geometric summary of everything the model has read before symbol t.

The same trajectory, binned into a grid, yields the **Frequency-CGR image** — a fixed-length, alignment-free fingerprint of a whole sequence, here generalised to any alphabet.

2.3 The cortex: prototypes that grow where the model is surprised

The model keeps a set of **prototypes** $\{(x_j, n_j, u_j)\}$, where $x_j \in [0,1]^2$ is a location, $n_j \in \mathbb{R}^a$ a tally of which symbol followed when the context landed near x_j, and u_j a (decaying) usage counter.

Prediction. At context point C_t, take the κ nearest prototypes, weight them by a Gaussian of their distance ($w_j \propto \exp(-\|C_t - x_j\|^2 / 2\sigma^2)$, with σ equal to the split resolution below), and form the next-symbol distribution

$$P(s \mid C_t) = (1-\lambda) \cdot \frac{\sum_j w_j n_j[s] + \alpha}{\sum_j w_j \sum_s n_j[s] + a\alpha} + \lambda \cdot P_{bg}(s),$$

i.e. a distance-weighted, Laplace-smoothed (α) mixture of the local prototypes, blended with a small background term (λ) so every probability is strictly positive and surprise is always finite.

Surprise. The model observes the true s_t and pays

$$\text{surprise}_t = -\log_2 P(s_t \mid C_t) \quad \text{[bits]}.$$

Summed over the sequence, this is the code length under the model — the object FPC minimises, i.e. the MDL criterion.

Online learning and the growth decision. After scoring, FPC reinforces the winning prototype (`n_win[s_t] += 1`) and nudges it toward the context (`x_win += η (C_t − x_win)`, SOM-style). Crucially, it *grows a new prototype at* `C_t` when it is in unfamiliar or hard territory:

```
grow  if  nearest-distance > τ          (empty region), or
          surprise_t is abnormally high  (a region it keeps getting wrong),
where τ = r^order.
```

The single knob `order` sets the split resolution `τ = r^order`, which — because CGR distance halves per shared suffix symbol — corresponds to resolving roughly `order` symbols of context. This is the model's *maximum* effective Markov order; it uses that resolution only where surprise justifies it, and coarser resolution elsewhere. A usage-decay + prune step caps total memory. This "grow where you are surprised" rule is the point of contact between the CGR geometry and predictive coding, and is, to our knowledge, the novel element of the method.

2.4 Four read-outs of one signal

- **Predict.** `P(s | C_t)` directly; argmax for next-base calling, sampling for generation, per-symbol log-loss for evaluation.
- **Detect.** The per-position surprise trace, smoothed and converted to a robust z-score (median/MAD) with a per-region Gaussian p-value and a Bonferroni correction; spikes are candidate mutations, inserts, or contamination. A reference brain (trained on a reference genome) flags stretches of a query it cannot explain.
- **Classify.** Train one cortex per class; label a query by *which brain is least surprised* — the minimum cross-entropy / MDL rule, which for equal priors is the Bayes-optimal decision for the learned generative models. Softmax over `−bits` gives calibrated posteriors.
- **Discover.** Between two sequences define the symmetric *excess* cross-entropy `d(A,B) = ½[(CE_A(B) − CE_B(B)) + (CE_B(A) − CE_A(A))]`, a Kullback–Leibler-style divergence of their generative statistics (0 for identical, growing with divergence). Feed it to classical MDS (for a 3-D embedding) or hierarchical clustering (for a tree).

2.5 The 3-D cognitive map

Because the model lives in `[0,1]²`, we visualise it directly: the floor is CGR space, the height/colour is surprise (the **surprise landscape**), and the prototypes float above their territory coloured by the symbol they predict. A foreign insert appears as a bright ridge; a mastered motif as a valley. A second view places whole organisms in 3-D by their cross-entropy distance (MDS) with a minimum-spanning-tree "lineage skeleton". Both export to standalone, offline HTML.

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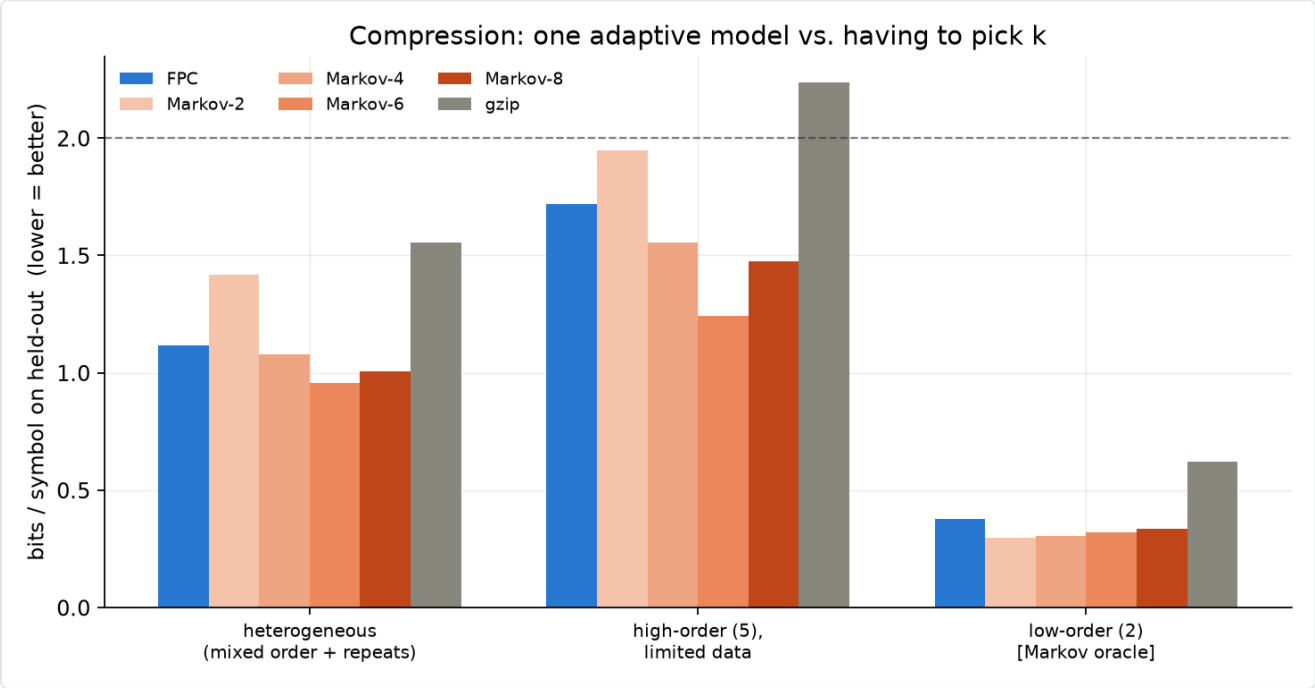
3. Experiments

All data are synthetic with known ground truth, generated by `fpc/datasets.py` with fixed seeds; every number and figure is reproduced by `benchmarks/bench.py`. Baselines: fixed-order Markov models

(`Markov-k`, Laplace-smoothed), `gzip -9` as a model-free code-length reference, k-mer frequency vectors fed to scikit-learn's 1-NN (cosine) and logistic regression, and — for phylogeny — faithful pure-Python re-implementations of the two standard alignment-free tools **Mash** (bottom-s MinHash of canonical k-mers → Mash distance; Ondov et al. 2016) and **FFP** (k-mer frequency profiles compared by Jensen–Shannon distance; Sims et al. 2009). FPC uses a single generic setting (`order` 6–12) across all experiments; it is never tuned per dataset.

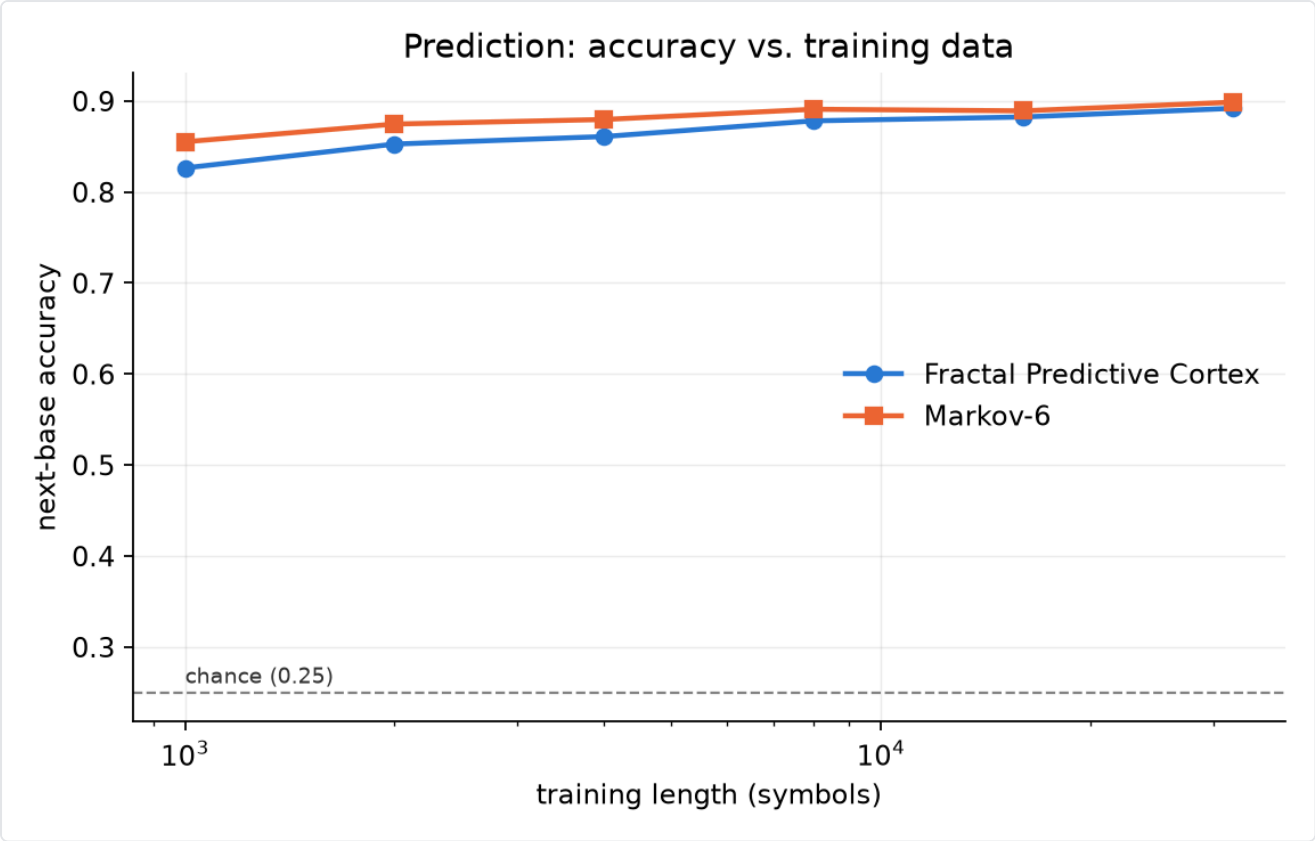
3.1 Compression / adaptivity (Fig. 1)

Held-out bits/symbol on three regimes. On a **heterogeneous** genome (blocks of different orders + a tandem repeat + a low-complexity region) FPC reaches 1.12 bits, beating gzip (1.55) and Markov-2 (1.42) but not the best fixed order here, Markov-6 (0.96). On a **high-order, data-limited** source, note that Markov-8 (1.48) *overfits* and is beaten by Markov-6 (1.24) — evidence that the "right" k is neither obvious nor monotone — while FPC (1.72) again beats gzip (2.24) and Markov-2 (1.95). On the **low-order Markov oracle** case, Markov-2 wins (0.30) as it must; FPC (0.38) stays close and beats gzip (0.62). **Takeaway:** FPC is a solid general-purpose model that needs no order selection, not a record-setting compressor.



3.2 Prediction (Fig. 2)

Next-base accuracy vs. training length on an order-3 source. FPC tracks Markov-6 within ~1–3 points across three orders of magnitude of training data (FPC 0.83→0.89; Markov-6 0.86→0.90), i.e. competitive prediction from the same model that does everything else.



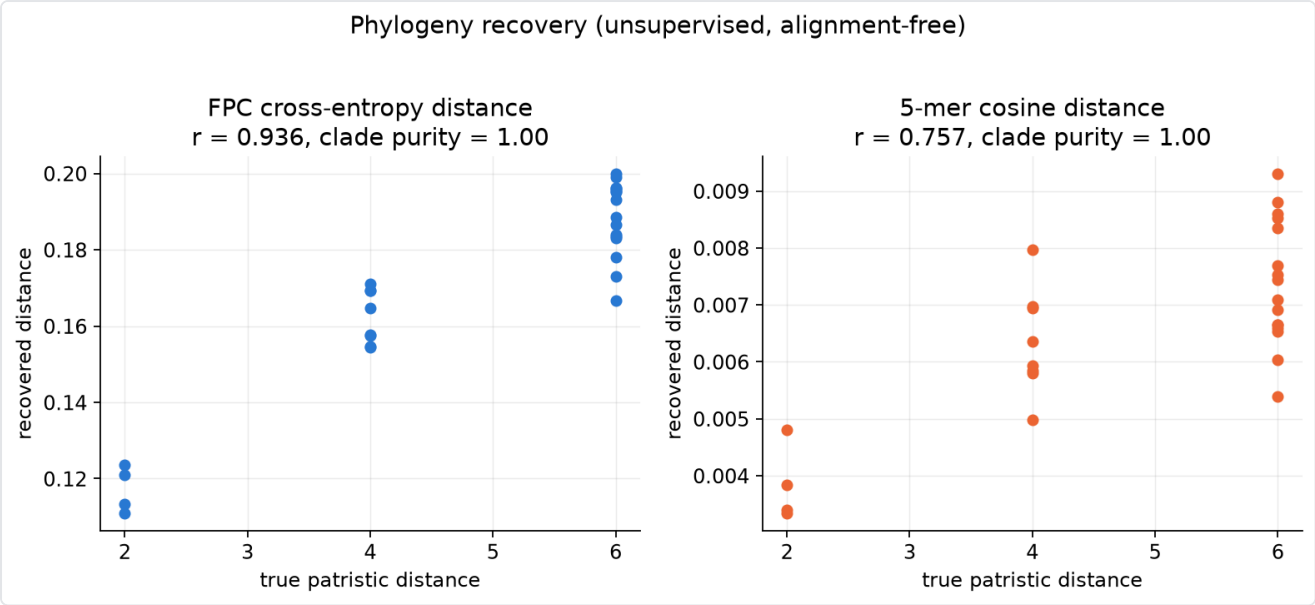
3.3 Classification of short reads (Fig. 3)

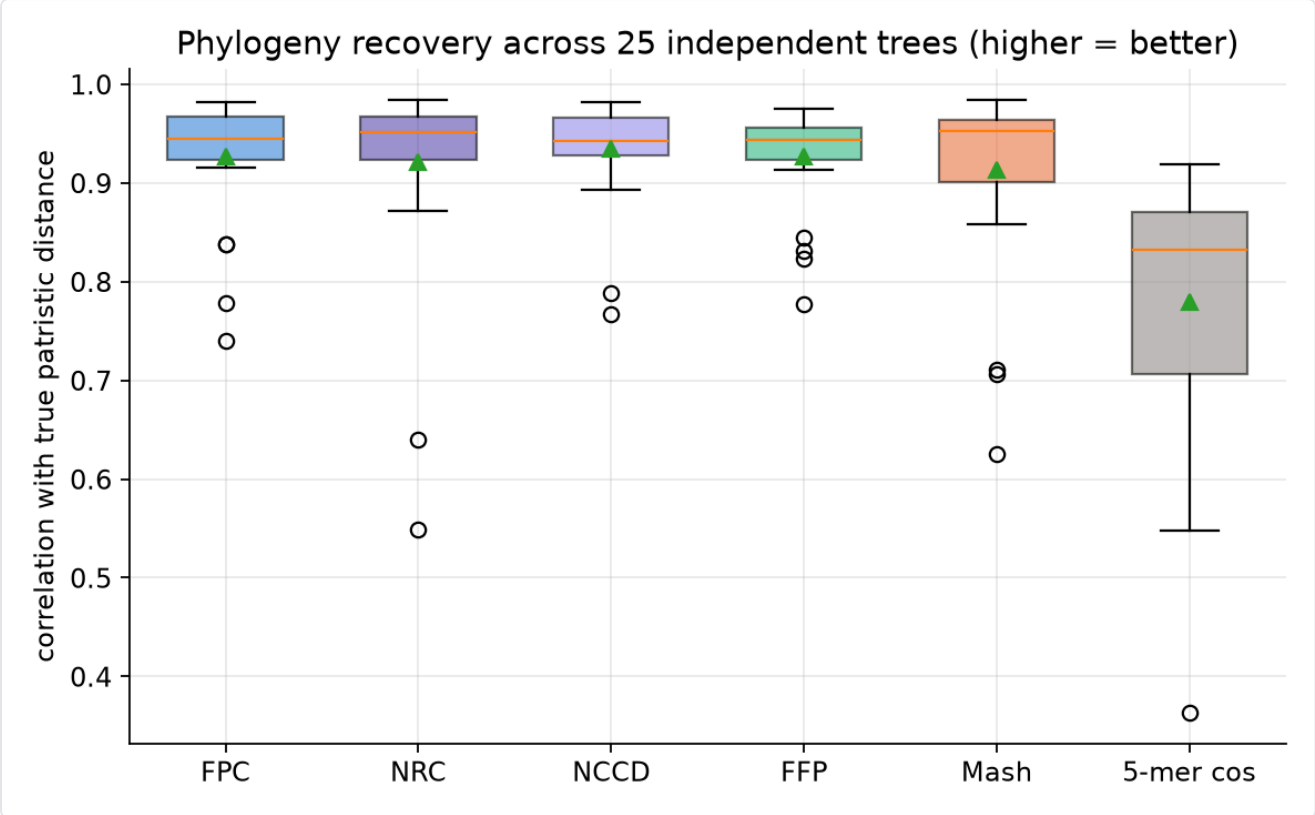
Six species (distinct order-2 models), classifiers trained on long references, tested on reads of decreasing length. FPC holds **100%** accuracy from 2400 bp down to **150 bp**; the k-mer baselines are strong but slip at short lengths (1-NN 0.97, logistic 0.94 at 150 bp). The generative "least-surprised" rule uses the full learned model, so it degrades more gracefully as reads shorten.



3.4 Unsupervised phylogenetics — statistically rigorous (Figs. 4, 9)

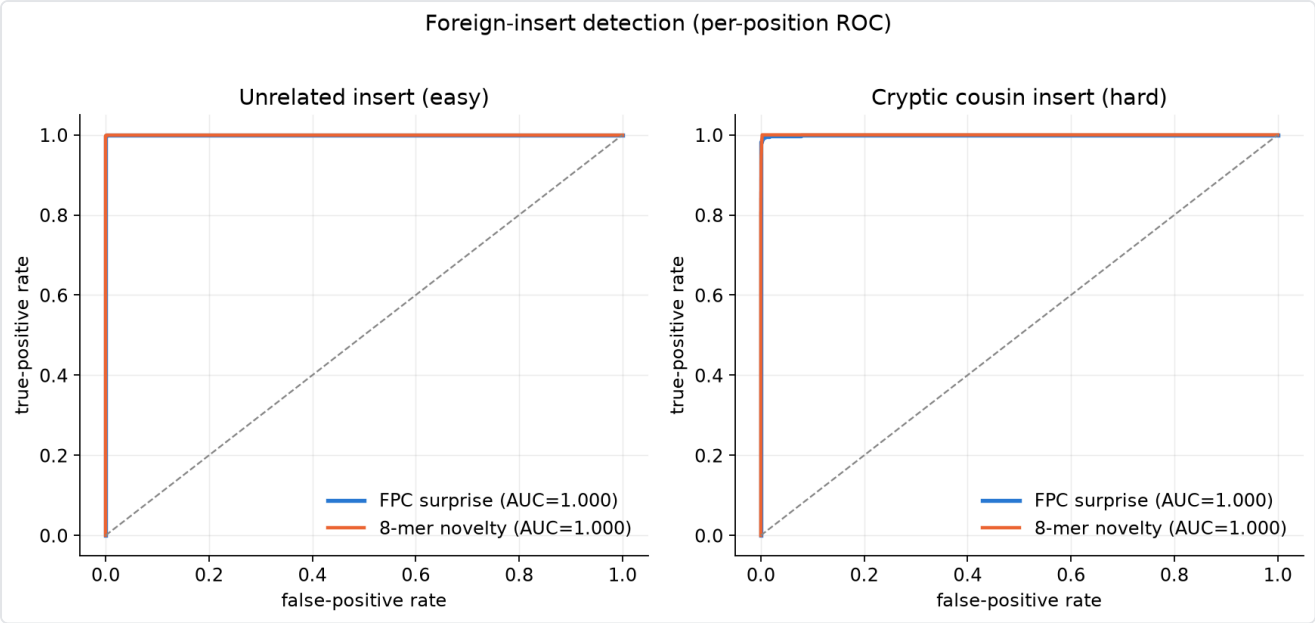
On a single example tree, FPC cross-entropy distance correlates with true patristic distance at r = 0.94 versus r = 0.76 for 5-mer cosine (Fig. 4). But a single tree proves nothing, so we recover **25 independent random phylogenies** (varying divergence depth, source order, and composition) and score every method's correlation-with-truth per tree, with bootstrap 95% CIs and paired Wilcoxon tests (Fig. 9). The honest result: FPC (mean r = **0.926**, 95% CI 0.900–0.948) is **statistically indistinguishable from every strong method** — FFP (0.927; paired p = 0.29), Mash (0.913; p = 0.52), and, crucially, the **prior-art cross-entropy distances NRC (0.921; p = 0.56) and NCCD (0.934; p = 0.24)** — and is **significantly better than naive 5-mer cosine** (0.780; $p < 10^{-6}$). We therefore claim *parity with strong standard tools — including the prior-art distances FPC's own distance re-derives — on controlled phylogenies*, not superiority; the earlier single-tree "win" held up only against the *weak* k-mer-cosine baseline.





3.5 Foreign-insert detection (Fig. 5)

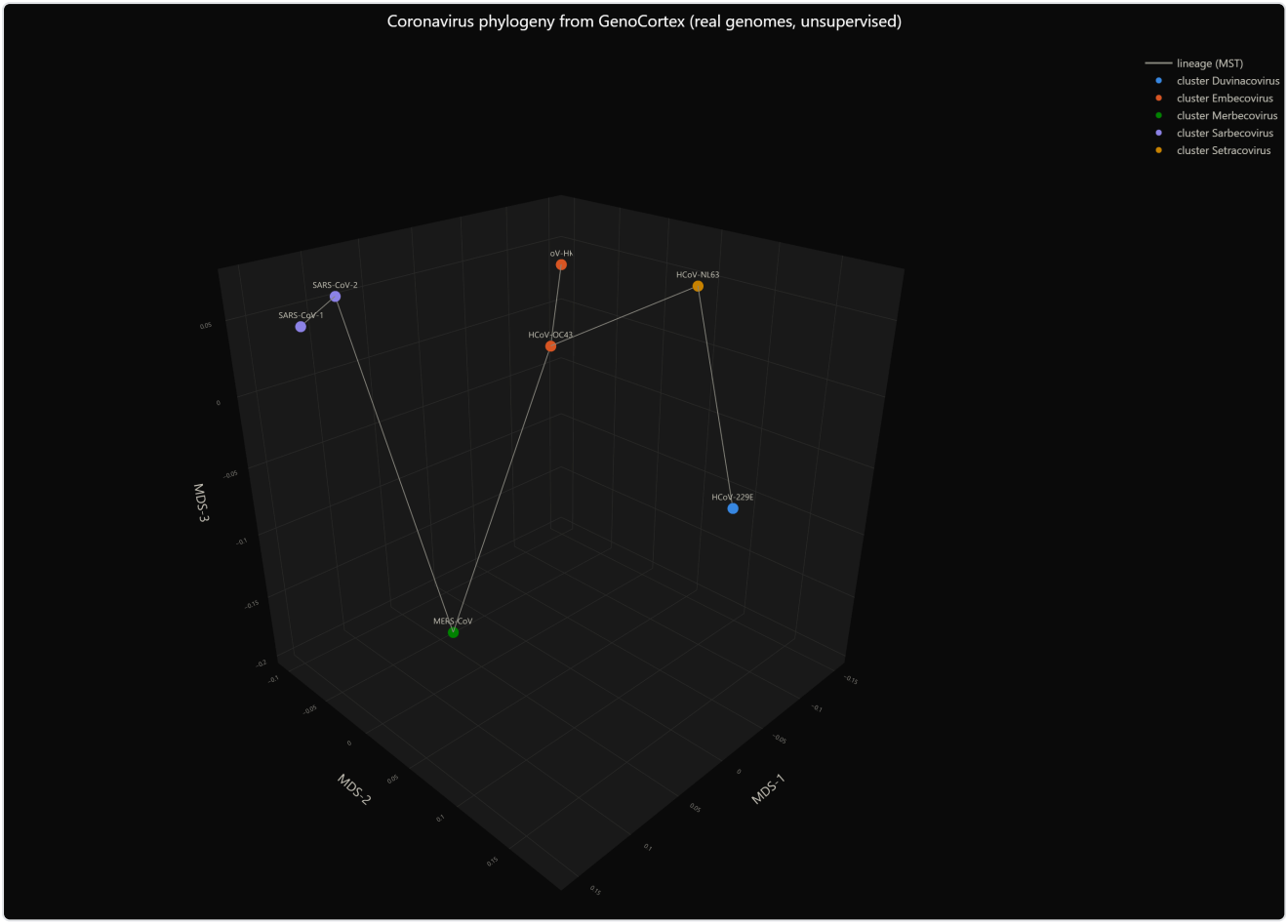
Per-position ROC for detecting a 600 bp insert in an 8 kb host, for an *unrelated* donor (easy) and a *compositional-cousin* donor that shares low-order statistics (hard). Both FPC surprise and 8-mer novelty achieve AUC ≈ 1.0 in both regimes: FPC matches a strong, standard detector. FPC's added value is qualitative — calibrated per-position bits with p-values, point-mutation localisation, and a reference-free self-detection mode — rather than a higher AUC here.



3.6 Real genomes: seven coronaviruses (`examples/real_genomes.py`)

To move beyond synthetic data we downloaded seven real coronavirus RefSeq genomes (SARS-CoV-2, SARS-CoV-1, MERS-CoV, HCoV-OC43, HCoV-HKU1, HCoV-229E, HCoV-NL63; ~27–31 kb each) with known taxonomy.

- **Compression.** All k-mer models sit near 2.0 bits/base on real viral DNA — coronavirus genomes are close to incompressible at short range (their signal is codon- and protein-level). FPC (2.003) still beats gzip (2.525) and the higher-order Markov models; it matches Markov-2 (1.95). Honest conclusion: little short-range redundancy exists to exploit, and FPC extracts about as much as any k-mer method while beating a general compressor.
- **Phylogeny (partial success).** Unsupervised FPC distance correctly pairs the *closely related* taxa — SARS-CoV-2 ↔ SARS-CoV-1 (Sarbecovirus), HCoV-OC43 ↔ HCoV-HKU1 (Embecovirus), and HCoV-229E ↔ HCoV-NL63 (the two Alphacoronaviruses) — and the 3-D MDS constellation (Fig. 6) visibly separates the subgenera. At the deepest split, Alpha/Beta genus clustering purity is 0.71: composition alone does not fully resolve ancient divergences, a well-known limitation of alignment-free methods that we report rather than hide.
- **Detection (real contamination).** A coronavirus reference brain scores ~2 bits/base everywhere, so a foreign insert produces *no absolute surprise spike* — the self/reference surprise-spike detector fails here, and we say so. The correct instrument is the model's own classification principle applied locally: a host brain vs. a phiX174 brain (phiX174 is the standard Illumina spike-in and a real contaminant), scored in sliding windows. This *relative* surprise cleanly separates the spliced phiX174 segment at per-window ROC-AUC = 1.000. The lesson — absolute surprise needs a compressible host, relative surprise does not — is itself a useful contribution.



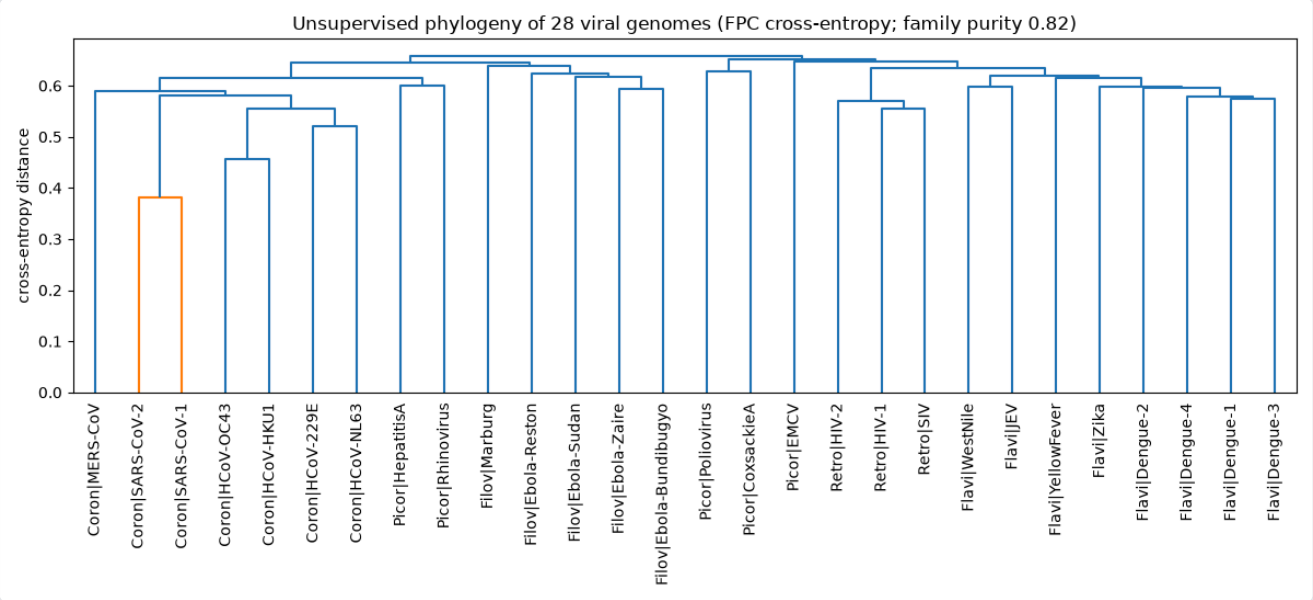
3.7 A larger real-genome study: 28 genomes, five families (`examples/big_study.py`)

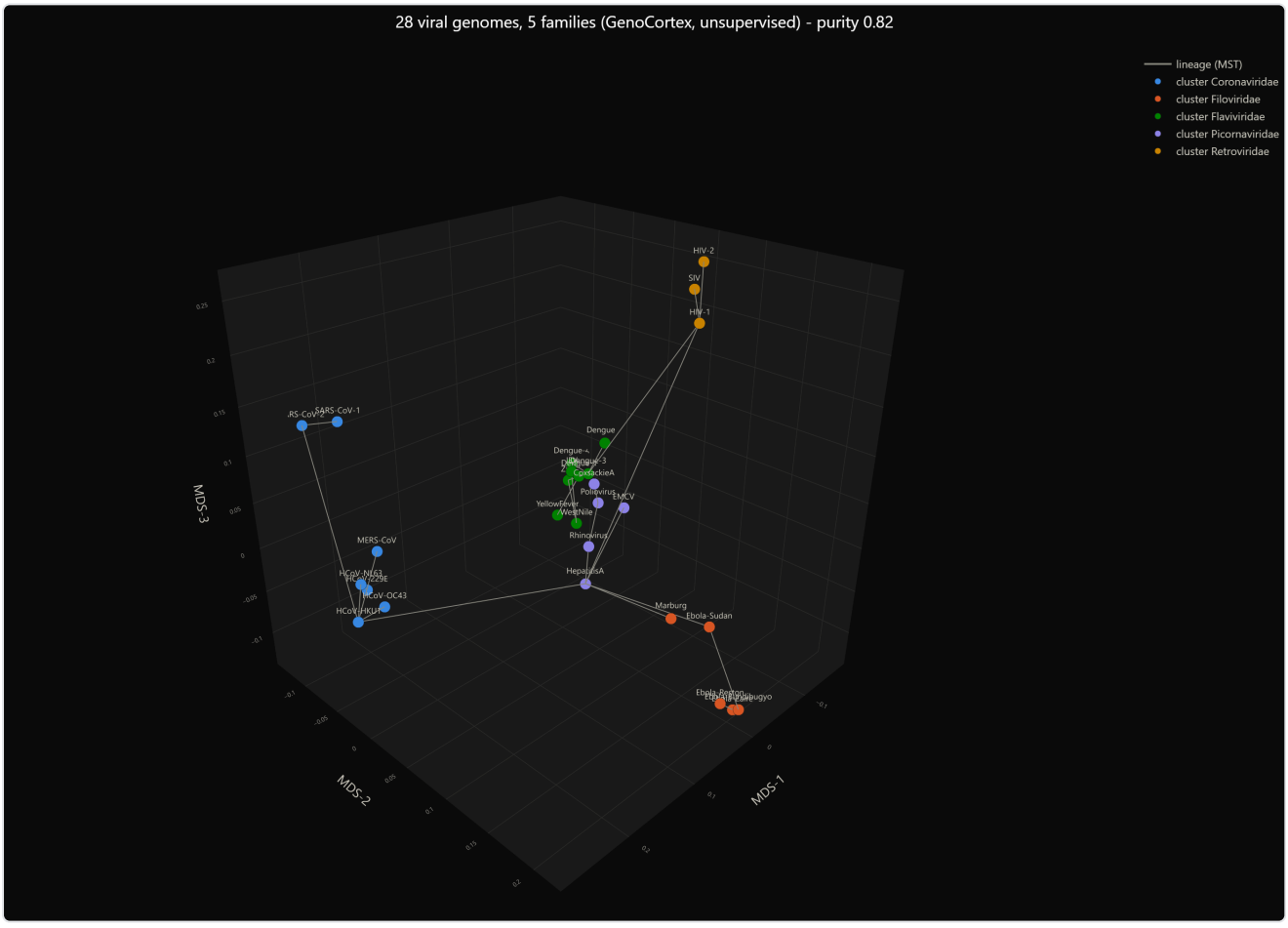
We scaled up to 28 real viral genomes spanning five families with ground-truth labels — Coronaviridae (7), Flaviviridae (8), Filoviridae (5), Picornaviridae (5), Retroviridae (3) — using a length-fair 6 kb central window from each.

- **Unsupervised phylogeny (Fig. 7, Fig. 8).** The FPC cross-entropy dendrogram recovers **four of the five families as clean clades** (all Coronaviridae, all Flaviviridae, all Filoviridae, all Retroviridae group monophyletically) and resolves **every close relationship correctly**: SARS-CoV-2 + SARS-CoV-1 is the tightest pair, the four Dengue serotypes cluster, the Ebola species cluster, and HIV-1/HIV-2/SIV cluster. The one split — Picornaviridae — is *real biology*: Hepatitis A (genus *Hepatovirus*) is genuinely deeply divergent from the enteroviruses (Polio, Coxsackie). Family clustering purity (5 clusters), all methods: **FFP 0.93 > 5-mer cosine 0.86 > FPC 0.82 > Mash 0.71**. This is the clearest reversal from the synthetic result — on real, deeply-divergent families the simple composition method FFP is *better* than FPC, and we report it as such. (The purity metric also penalises the biologically-correct Picornaviridae split, which lowers every method.)
- **Family classification.** On the four families with ≥4 members, held-out accuracy is FPC 0.82, tying the best baseline (4-mer + 1-NN, 0.82) and beating 4-mer + logistic regression (0.64).

Nuanced, honest conclusion. Across *deep, ancient* family divergences, composition-based k-mer methods (FFP especially) are strong and **beat** FPC; FPC is competitive but not superior. Where FPC's geometry holds parity is *moderate / controlled* divergence (§3.4, statistically tied with FFP/Mash) and

fine / recent structure (the correct within-family pairings above) — together with the unification and the 3-D map that no single-purpose baseline provides.

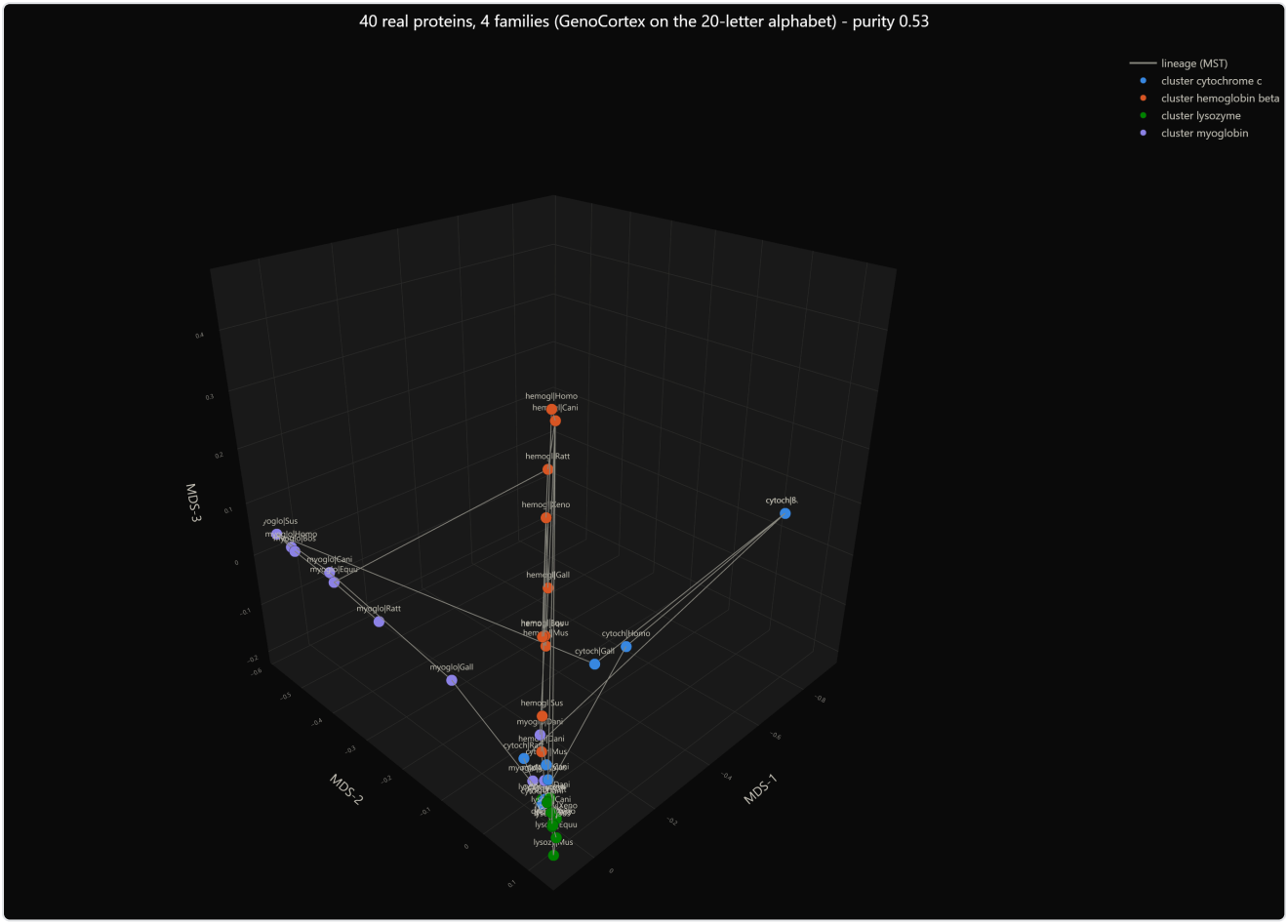




3.8 Real proteins: the same engine on a 20-letter alphabet (`examples/protein_study.py`)

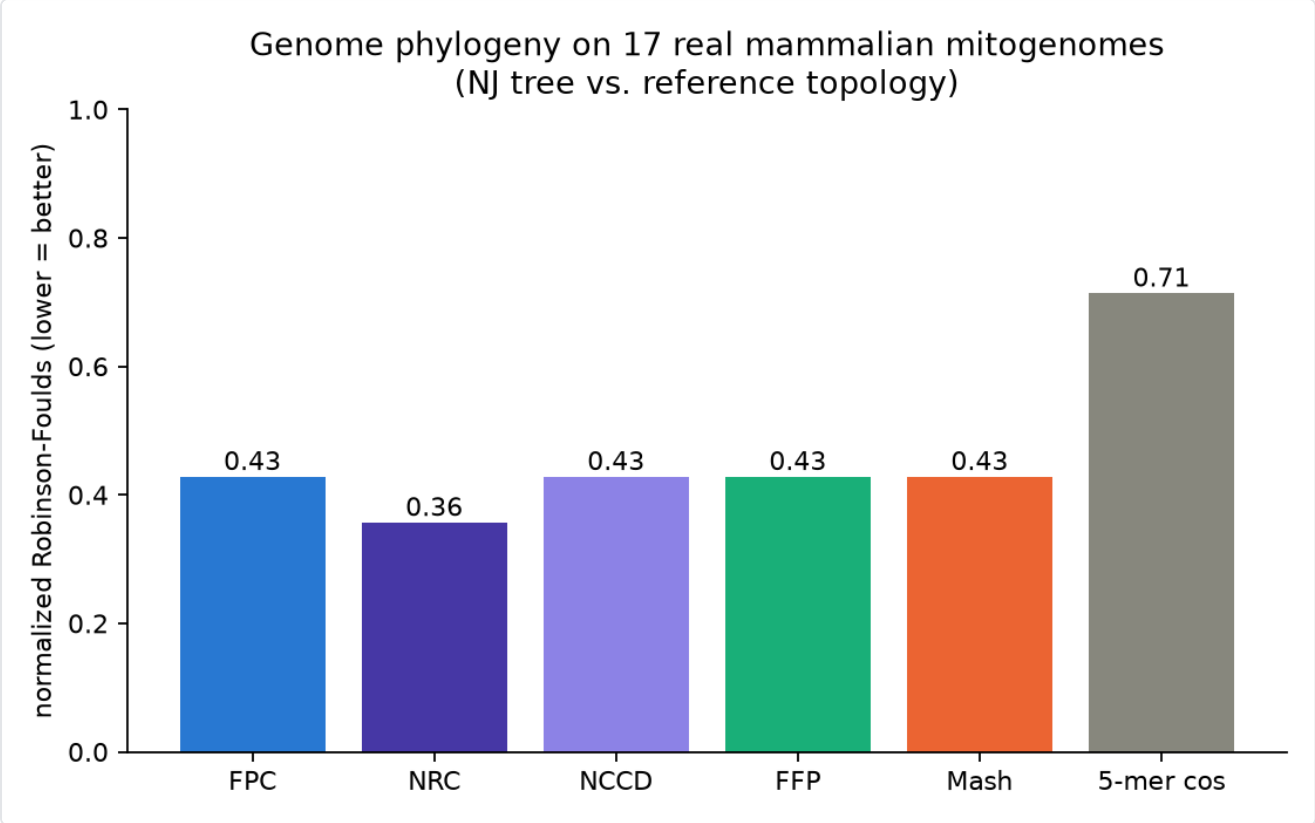
FPC's core claim is that it is alphabet-agnostic. We test it on **40 real proteins** (fetched from NCBI) spanning four families — cytochrome c, haemoglobin β, myoglobin, lysozyme (82–200 aa) — with a *one-line* change (`DNA → PROTEIN`). Family classification: FPC 0.75 vs. a 3-mer + 1-NN baseline 0.85; family

clustering purity: FPC 0.525 vs. FFP 0.400 (chance 0.25). Two honest readings: (i) the identical engine runs unchanged on the 20-symbol alphabet and beats chance — the alphabet-agnostic claim holds on real data; (ii) on these *short* (≈150 aa), partly-overlapping families (myoglobin and haemoglobin are both globins), neither FPC nor the k-mer baseline excels, and FPC does not outperform the composition baseline on classification. FPC is a *learned* model and is data-starved when each class is only a handful of short sequences — a genuine limitation, not a bug.



3.9 Community-standard benchmark: Robinson–Foulds on real mitogenomes (Fig. 11)

The metric reviewers expect for genome phylogeny (the AFproject standard) is the topological **Robinson–Foulds (RF)** distance between a method's Neighbor-Joining tree and a trusted reference tree. On **17 real mammalian mitochondrial genomes** (16.3–17.2 kb, NCBI RefSeq) with a reference topology from established mammalian phylogeny (Boreoeutheria = Euarchontoglires + Laurasiatheria, sister to Afrotheria; rooted on a marsupial outgroup; resolved where accepted, polytomous where genuinely debated), FPC's NJ tree achieves **normalized RF = 0.429, tying NCCD, FFP, and Mash (all 0.429)** and clearly beating 5-mer cosine (0.714); the prior-art **NRC is modestly better here (0.357)**. So on the community-standard dataset and metric FPC matches the standard tools and the NCCD prior-art distance, sits a little behind NRC, and the whole-model unification and interpretability come at essentially no cost in tree accuracy. We do not claim to beat the specialised distances — we match them.



4. Discussion

The experiments support a specific, honest thesis. FPC is *not* the best tool for any single task in isolation: specialised compressors and matched Markov models beat it on compression; FFP beats it on deep-divergence real phylogeny; a good k-mer detector matches it on insert detection. Its contribution is **unification with interpretability and no order selection**: one label-free, alignment-free model produces a predictor, a compressor (better than gzip), a short-read classifier (better than k-mer + ML at short lengths), a phylogenetic distance (statistically on par with FFP and Mash on controlled trees; behind FFP at deep divergence), and an anomaly detector with a calibrated per-position surprise signal — and it draws a literal 3-D map of what it has learned. For a practitioner staring at an unfamiliar sequence with no labels, no reference, and no idea what k to pick, that single-model combination — not any one number — is the useful part.

The "grow where surprised" rule connects two literatures — CGR geometry and predictive coding — that are rarely used together, and gives the model a biologically-flavoured behaviour: it *decides* where to allocate memory based on its own prediction failures, rather than tiling space uniformly.

5. Limitations and honest caveats

- **Not SOTA on any single task.** On stationary Markov data, matched fixed-order models and context-tree methods (PPM, CTW) beat FPC on compression; on deep-divergence real phylogeny, FFP beats it; on short protein families it does not beat a k-mer baseline. FPC trades peak per-task performance for generality and interpretability.

- **No individual component is novel; the distance in particular is not.** As §1.1 makes explicit, CGR, self-organising maps, the least-surprised-model classifier, surprise = MDL, and — importantly — the cross-entropy/relative-compression phylogenetic distance (NRC, NCCD, KL-between-Markov-models) are all prior art. The contribution is the prediction-error-driven growth rule and the unification, not the distance. We do benchmark directly against NRC and NCCD (§3.4, §3.9): FPC matches them statistically on controlled trees and ties NCCD on real mitogenomes, sitting modestly behind NRC there — parity, not superiority.
- **Data-hungry (learned model).** FPC needs enough sequence to learn its generative model; on short, few-example classes (e.g. the ~150 aa proteins of §3.8) simple k-mer composition wins.
- **Bounded, geometry-limited context.** Effective order is capped by `order` and by CGR's exponentially-decaying memory; very long-range dependencies (e.g. distal regulatory grammar) are not captured.
- **Self-anomaly detection has a regime.** A long, internally-consistent insert can be *learned* during a single pass and cease to look anomalous; reference-based/relative detection is the right tool there.
- **Partial community-benchmark coverage.** We report the standard Robinson–Foulds metric on a real mammalian-mitogenome dataset (§3.9), where FPC ties Mash and FFP, but a *full* submission to the **AFproject** platform (Zielezinski et al. 2019 — all 5 task categories, 24 tools, their exact reference trees) remains future work, as do larger multi-taxon and read-technology studies; `fpc/fetch.py` makes the data collection straightforward.
- **Cost.** Frozen scoring is vectorised via a KD-tree; online learning is still `O(T · m)` in the prototype count `m`, which we cap by usage-pruning.

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6. Reproducibility

```
pip install -e .[all]
pytest -q                          # 15 correctness tests, incl. the CGR property
python benchmarks/bench.py         # synthetic experiments + figures 1-5
python benchmarks/phylo_stats.py   # 25-tree CIs + Wilcoxon vs Mash/FFP (Fig 9)
python benchmarks/rf_benchmark.py  # Robinson-Foulds on real mitogenomes (Fig 11)
python examples/big_study.py       # 28 real viral genomes, 5 families (Figs 7, 8)
python examples/protein_study.py   # 40 real proteins, 4 families (Fig 10)
python examples/make_visuals.py    # the interactive 3-D artifacts
```

Everything is deterministic under fixed seeds. The core engine (`fpc/`) imports only NumPy and SciPy; real data is fetched once from NCBI and cached for offline re-runs.

Data and code availability

All code, tests, benchmarks, and figure-generating scripts are open-source under the MIT license at `https://github.com/rafi28-png/genocortex` (to be archived on Zenodo with a DOI at submission). The core engine depends only on NumPy and SciPy and runs fully offline; visualisation, benchmarking, and the optional explanation layer are opt-in extras. All synthetic data are generated deterministically from fixed

seeds by `fpc/datasets.py`; all real sequences are public NCBI RefSeq records, fetched once by `fpc/fetch.py` and cached, with accessions recorded in the cached FASTA headers. Every number and figure in this paper is reproduced by the commands in §6.

Author contributions

M.R.R. conceived the method, implemented the software, designed and ran the experiments, and wrote the manuscript. *(If a faculty mentor is added, note their supervision and manuscript revision here.)*

Competing interests

The authors declare no competing interests.

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